

```
from openai import OpenAI

client = OpenAI(
    base_url="https://openrouter.ai/api/v1",
    api_key="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",
)

completion = client.chat.completions.create(
    extra_headers={
        "HTTP-Referer": "<YOUR_SITE_URL>", # Optional. Site URL for rankings on openrouter.ai.
        "X-Title": "<YOUR_SITE_NAME>", # Optional. Site title for rankings on openrouter.ai.
    },
    extra_body={},
    model="deepseek/deepseek-r1",
    messages=[
        {
            "role": "user",
            "content": "how non technical learner transit towards datasciece?"
        }
    ]
)

print(completion.choices[0].message.content)
```

Transitioning into data science from a non-technical background is a challenging but achievable journey with a structured approach.

- ```
1. **Build Foundational Knowledge**
- **Statistics & Math**: Start with basic statistics (mean, median, variance, hypothesis testing) and probability. Use resources like Khan Academy.
- **Intro to Data Science**: Understand the data lifecycle (collection, cleaning, analysis, modeling) through free resources like DataCamp's "Data Science Fundamentals".

2. **Learn Programming Basics**
- **Python**: Begin with Python fundamentals (syntax, loops, functions). Use interactive platforms like Codecademy or "Python for Data Science".
- **SQL**: Learn to query databases with platforms like SQLZoo or Mode Analytics. Focus on SELECT, JOIN, and aggregation.

3. **Data Manipulation & Visualization**
- **Pandas/NumPy**: Master data cleaning and analysis using Python libraries. Try Kaggle micro-courses or "Python for Data Analysis".
- **Visualization**: Use Matplotlib/Seaborn (Python) or Tableau Public. Practice with datasets from Kaggle or Google Data Science.

4. **Machine Learning (ML) Fundamentals**
- **Concepts**: Study ML basics (supervised vs. unsupervised learning, regression, classification). Andrew Ng's ML course is a great starting point.
- **Libraries**: Implement algorithms with scikit-learn. Start with simple projects (e.g., predicting house prices).

5. **Projects & Portfolio**
- **Beginner Projects**: Analyze datasets (e.g., Titanic survival, COVID trends). Showcase work on GitHub and write blog posts.
- **Advanced Projects**: Tackle Kaggle competitions or build end-to-end projects (e.g., customer churn prediction).

6. **Networking & Soft Skills**
- **Communities**: Join r/datascience, LinkedIn groups, or local meetups. Attend webinars/hackathons.
- **Communication**: Practice presenting insights using tools like PowerPoint or Tableau. Write clear documentation for projects.

7. **Continuous Learning & Job Prep**
- **Stay Updated**: Follow blogs (Towards Data Science) or podcasts (Data Skeptic).
- **Resume & Interviews**: Highlight projects and skills. Prepare for technical interviews with platforms like LeetCode (for coding) and DataCamp (for data science).
```

#### ### Tools & Resources Cheat Sheet

| **Area**          | **Tools/Resources**                                             |
|-------------------|-----------------------------------------------------------------|
| **Python Basics** | Codecademy, "Automate the Boring Stuff" (book)                  |
| **SQL**           | SQLZoo, LeetCode, Mode Analytics                                |
| **Data Analysis** | Pandas, Kaggle micro-courses, "Python for Data Analysis" (book) |
| **Visualization** | Matplotlib/Seaborn, Tableau Public, DataCamp                    |
| **ML**            | Andrew Ng's ML course, scikit-learn, Kaggle Learn               |
| **Projects**      | GitHub, Jupyter Notebooks, Google Colab                         |
| **Communities**   | Reddit/r/datascience, Data Science Discord, Meetup.com          |

#### ### Example Learning Timeline

- \*\*Months 1-2\*\*: Python + Statistics basics.
- \*\*Months 3-4\*\*: SQL + Pandas/NumPy + Visualization.
- \*\*Months 5-6\*\*: ML fundamentals + 2-3 projects.
- \*\*Months 7+\*\*: Advanced projects, portfolio building, and networking.

#### ### Key Tips

- \*\*Tutorial Hell\*\*: Avoid passive learning—apply concepts immediately.
- \*\*Consistency\*\*: Dedicate 5-10 hours/week; use apps like Toggl to track time.
- \*\*Mindset\*\*: Embrace mistakes as learning opportunities. Imposter syndrome is common—focus on progress, not perfection.

By breaking the journey into manageable steps and staying persistent, you'll gradually build the technical and analytical skills needed for a career in data science.

```
Install required libraries installed
!pip install streamlit ollama

import streamlit as st
import ollama

Page config
```

```

st.set_page_config(
 page_title="Kimi 2.5 Chat (Ollama)",
 page_icon="🐙",
 layout="centered"
)

st.title("🐙 Kimi 2.5 - Local Chat")
st.caption("Powered by Ollama • Fully Offline • Streaming")

Session state for chat history
if "messages" not in st.session_state:
 st.session_state.messages = []

Display chat history
for msg in st.session_state.messages:
 with st.chat_message(msg["role"]):
 st.markdown(msg["content"])

User input
prompt = st.chat_input("Ask something...")

if prompt:
 # Store user message
 st.session_state.messages.append({
 "role": "user",
 "content": prompt
 })

 with st.chat_message("user"):
 st.markdown(prompt)

 # Assistant response (streaming)
 with st.chat_message("assistant"):
 placeholder = st.empty()
 full_response = ""

 stream = ollama.chat(
 model="kimi-k2.5:cloud",
 messages=st.session_state.messages,
 stream=True
)

 for chunk in stream:
 if "message" in chunk and "content" in chunk["message"]:
 token = chunk["message"]["content"]
 full_response += token
 placeholder.markdown(full_response)

 # Save assistant response
 st.session_state.messages.append({
 "role": "assistant",
 "content": full_response
 })

```

```

Collecting streamlit
 Downloading streamlit-1.58.0-py3-none-any.whl.metadata (9.6 kB)
Collecting ollama
 Downloading ollama-0.6.2-py3-none-any.whl.metadata (5.8 kB)
Requirement already satisfied: altair!=5.4.0,!5.4.1,<7,>=4.0 in /usr/local/lib/python3.12/dist-packages (from streamlit) (1.9.0)
Requirement already satisfied: blinker<2,>=1.5.0 in /usr/local/lib/python3.12/dist-packages (from streamlit) (1.9.0)
Requirement already satisfied: cachetools<8,>=5.5 in /usr/local/lib/python3.12/dist-packages (from streamlit) (6.2.6)
Requirement already satisfied: click<9,>=7.0 in /usr/local/lib/python3.12/dist-packages (from streamlit) (8.4.1)
Requirement already satisfied: gitpython!=3.1.19,<4,>=3.0.7 in /usr/local/lib/python3.12/dist-packages (from streamlit) (3.1.19)
Requirement already satisfied: numpy<3,>=1.23 in /usr/local/lib/python3.12/dist-packages (from streamlit) (2.0.2)
Requirement already satisfied: packaging>=20 in /usr/local/lib/python3.12/dist-packages (from streamlit) (26.2)
Requirement already satisfied: pandas<4,>=1.4.0 in /usr/local/lib/python3.12/dist-packages (from streamlit) (2.2.2)
Requirement already satisfied: pillow<13,>=7.1.0 in /usr/local/lib/python3.12/dist-packages (from streamlit) (11.3.0)
Collecting pydeck<1,>=0.8.0b4 (from streamlit)
 Downloading pydeck-0.9.2-py2.py3-none-any.whl.metadata (4.2 kB)
Requirement already satisfied: protobuf<8,>=3.20 in /usr/local/lib/python3.12/dist-packages (from streamlit) (5.29.6)
Requirement already satisfied: pyarrow>=7.0 in /usr/local/lib/python3.12/dist-packages (from streamlit) (18.1.0)
Requirement already satisfied: requests<3,>=2.27 in /usr/local/lib/python3.12/dist-packages (from streamlit) (2.32.4)
Requirement already satisfied: tenacity<10,>=8.1.0 in /usr/local/lib/python3.12/dist-packages (from streamlit) (9.1.4)
Requirement already satisfied: toml<2,>=0.10.1 in /usr/local/lib/python3.12/dist-packages (from streamlit) (0.10.2)
Requirement already satisfied: typing-extensions<5,>=4.10.0 in /usr/local/lib/python3.12/dist-packages (from streamlit) (4.12.0)
Requirement already satisfied: starlette>=0.40.0 in /usr/local/lib/python3.12/dist-packages (from streamlit) (0.52.1)
Requirement already satisfied: uvicorn>=0.30.0 in /usr/local/lib/python3.12/dist-packages (from streamlit) (0.49.0)
Requirement already satisfied: httpx>=0.6.3 in /usr/local/lib/python3.12/dist-packages (from streamlit) (0.8.0)
Requirement already satisfied: anyio>=4.0.0 in /usr/local/lib/python3.12/dist-packages (from streamlit) (4.13.0)
Requirement already satisfied: python-multipart>=0.0.10 in /usr/local/lib/python3.12/dist-packages (from streamlit) (0.0.3)
Requirement already satisfied: websockets>=12.0.0 in /usr/local/lib/python3.12/dist-packages (from streamlit) (15.0.1)
Requirement already satisfied: itsdangerous>=2.1.2 in /usr/local/lib/python3.12/dist-packages (from streamlit) (2.2.0)
Requirement already satisfied: watchdog<7,>=2.1.5 in /usr/local/lib/python3.12/dist-packages (from streamlit) (6.0.0)
Requirement already satisfied: httpx>=0.27 in /usr/local/lib/python3.12/dist-packages (from ollama) (0.28.1)
Requirement already satisfied: pydantic>=2.9 in /usr/local/lib/python3.12/dist-packages (from ollama) (2.12.3)

```

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Requirement already satisfied: Jinja2 in /usr/local/lib/python3.12/dist-packages (from altair!=5.4.0,!=5.4.1,<7,>=4.0->streamlit)
Requirement already satisfied: jsonschema>=3.0 in /usr/local/lib/python3.12/dist-packages (from altair!=5.4.0,!=5.4.1,<7,>=4.0->streamlit)
Requirement already satisfied: narwhals>=1.14.2 in /usr/local/lib/python3.12/dist-packages (from altair!=5.4.0,!=5.4.1,<7,>=4.0->streamlit)
Requirement already satisfied: idna>=2.8 in /usr/local/lib/python3.12/dist-packages (from anyio>=4.0.0->streamlit) (3.18)
Requirement already satisfied: gitdb<5,>=4.0.1 in /usr/local/lib/python3.12/dist-packages (from gitpython!=3.1.19,<4,>=3.0->streamlit)
Requirement already satisfied: certifi in /usr/local/lib/python3.12/dist-packages (from httpx>=0.27->ollama) (2026.5.20)
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (from httpx>=0.27->ollama) (1.0.9)
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx>=0.27->ollama) (0.14.0)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas<4,>=1.4.0->streamlit)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas<4,>=1.4.0->streamlit)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas<4,>=1.4.0->streamlit)
Requirement already satisfied: annotated-types>=0.6.0 in /usr/local/lib/python3.12/dist-packages (from pydantic>=2.9->ollama)
Requirement already satisfied: pydantic-core==2.41.4 in /usr/local/lib/python3.12/dist-packages (from pydantic>=2.9->ollama)
Requirement already satisfied: typing-inspection>=0.4.2 in /usr/local/lib/python3.12/dist-packages (from pydantic>=2.9->ollama)
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests<3,>=2.27->streamlit)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests<3,>=2.27->streamlit)
Requirement already satisfied: smmap<6,>=3.0.1 in /usr/local/lib/python3.12/dist-packages (from gitdb<5,>=4.0.1->gitpython)
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from Jinja2->altair!=5.4.0,!=5.4.1,<7,>=4.0->streamlit)
Requirement already satisfied: attrs>=22.2.0 in /usr/local/lib/python3.12/dist-packages (from jsonschema>=3.0->altair!=5.4.0,!=5.4.1,<7,>=4.0->streamlit)
Requirement already satisfied: jsonschema-specifications>=2023.03.6 in /usr/local/lib/python3.12/dist-packages (from jsonschema>=3.0->altair!=5.4.0,!=5.4.1,<7,>=4.0->streamlit)
Requirement already satisfied: referencing>=0.28.4 in /usr/local/lib/python3.12/dist-packages (from jsonschema>=3.0->altair!=5.4.0,!=5.4.1,<7,>=4.0->streamlit)
Requirement already satisfied: rpds-py>=0.25.0 in /usr/local/lib/python3.12/dist-packages (from jsonschema>=3.0->altair!=5.4.0,!=5.4.1,<7,>=4.0->streamlit)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas<4,>=1.4.0->streamlit)
Downloading streamlit-1.58.0-py3-none-any.whl (9.2 MB)

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9.2/9.2 MB 58.5 MB/s eta 0:00:00
Downloading ollama-0.6.2-py3-none-any.whl (15 kB)

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